Image Model

- Image refers to a 2-D light intensity function denoted by $f(x,y)$.
- At spatial location (x_0,y_0), f gives the intensity (Brightness) of the image at that point. $f(x_0,y_0)$ is called the **gray level** (l)
- It is evident that l lies in the range L_{max} L_{min}

$$f(x,y) = i(x,y)r(x,y)$$

where

$$0 < i(x,y) < \infty \quad \text{(illumination)}$$

and

$$0 < r(x,y) < 1 \quad \text{(reflectance)}$$

Image Model

- In practice, $L_{min} = i_{min} \, r_{min}$ and $L_{max} = i_{max} \, r_{max}$
- The interval $[L_{min}, L_{max}]$ is called the *gray scale*.
- It is common to shift this interval numerically to the interval (L levels): $[0, L-1]$, Where

$L = 0$ is considered black

$L = L-1$ is considered white in the gray scale

Digital Image

- Digital image is an analog image *f(x,y)* that has been discretized in
  - Space
  - Brightness
- *f(x,y)* can be
  - scalar function representing a monochrome image
  - vector valued function representing a colored image
- Each element of *f(x,y)* is called ***Pel*** or ***Pixel*** (Picture Element .)

Sampling and Quantization

- Sampling is digitizing the x and y coordinates (N×M points)
- Quantization is amplitude digitization into grey levels, normally, powers of 2
- Gray levels = 2^k, Number of bits = $N \times M \times k$

$$f(i,j)=\begin{bmatrix} f(0,0) & f(0,1) & .\ . & f(0,M-1) \\ f(1,0) & f(1,1) & .\ . & f(1,M-1) \\ . & & & \\ . & & & \\ f(N-1,0) & f(N-1,1) & .\ . & f(N-1,M-1) \end{bmatrix}$$

Image Samples

- Generally, image is of size m x n
  - where,
    - m: number of rows and
    - n: number of columns
- The no. of gray level values (L), is taken as power of 2
- $L = 2^k$
  - where,
    - k: number of bits used for representation

Image Size

- Eg:
  - For an image of 512 by 512 pixels, with 8 bits per pixel

Solution

$$
\begin{aligned}
\text{Size} &= 512 * 512 * 8 \text{ bits} \\
&= 2^9 * 2^9 * 2^3 \text{ bits} \\
&= 2^{21}/2^3 \text{ bytes} \\
&= 2^{18}/2^{10} \text{ K bytes} \\
&= 256 \text{ KB}
\end{aligned}
$$

Image Digitization

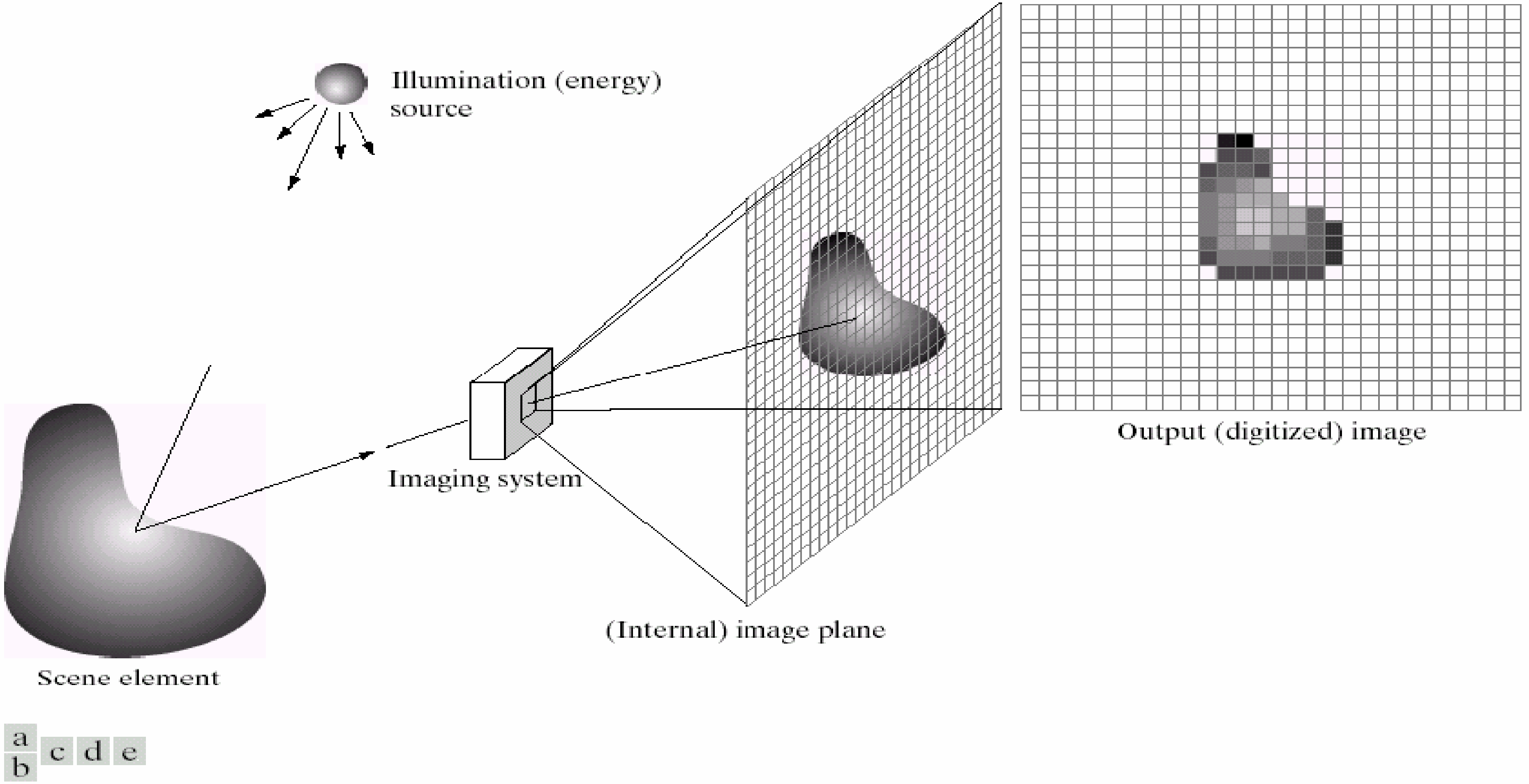
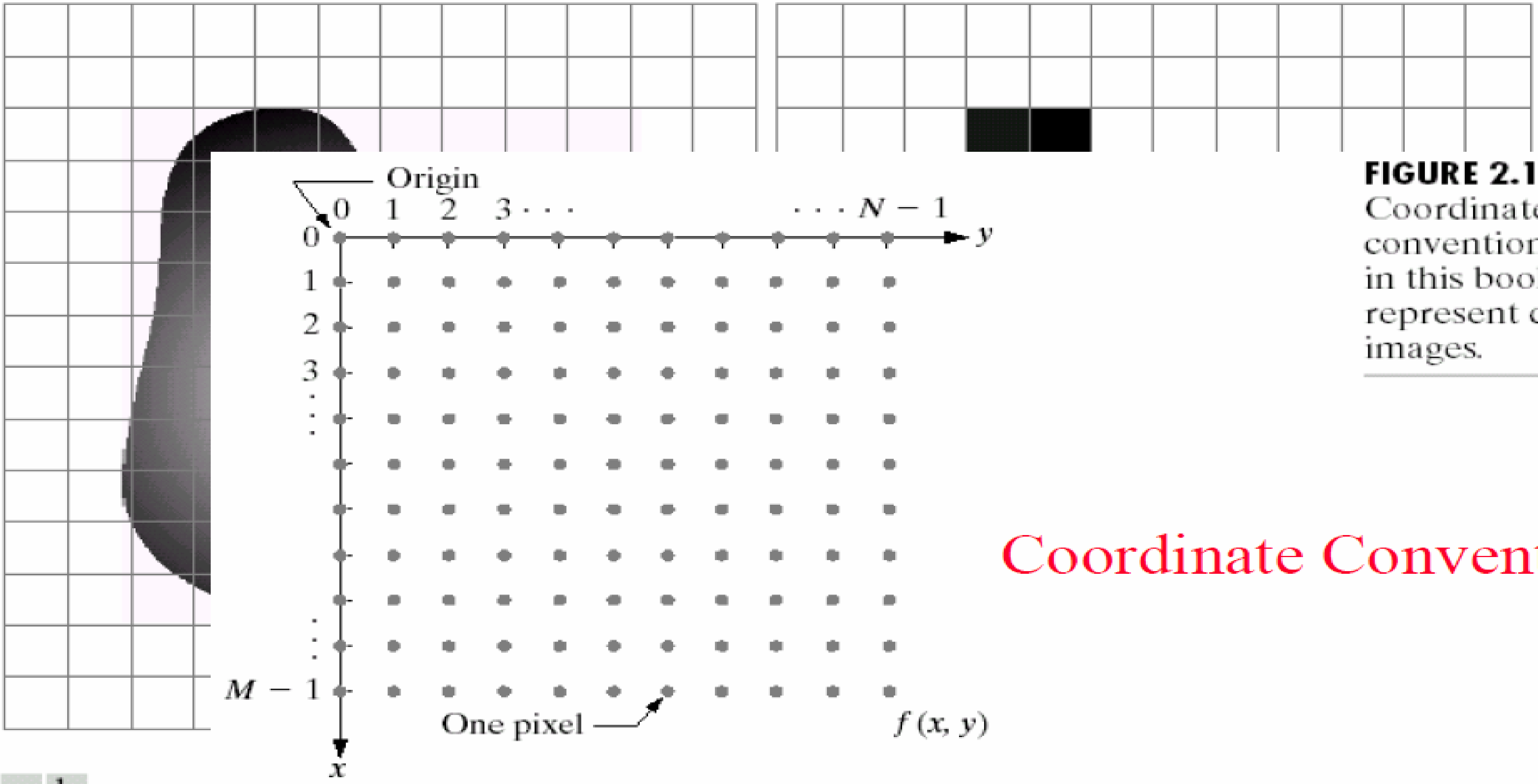


FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

Image Digitization



a b

FIGURE 2.17 (a) Continuos image projected onto a sensor array. (b) Result of image sampling and quantization.

FIGURE 2.18 Coordinate convention used in this book to represent digital images.

Coordinate Convention

This is a 256 × 256 digital image.

Picture Elements

11	11	11	11	12	12	12	11	10	11
12	12	13	13	13	14	13	13	14	14
13	12	12	12	12	13	14	15	15	13
13	12	12	12	13	15	14	15	14	12
12	11	12	13	16	15	14	15	13	12
12	12	13	14	16	15	16	13	12	12
12	14	16	15	15	14	14	13	12	12
13	16	15	14	13	13	14	13	13	12
12	14	13	12	12	13	13	13	12	11
11	11	11	11	12	15	15	13	11	8

Picture Elements (*pixels*) in grey-level values to be represented as an 8-bit unsigned integers

Gray levels = 2^k

Number of bits = N×M×k

For a square image= $N^2 \times k$

TABLE 2.1
Number of storage bits for various values of N and k.

N/k	1 (L = 2)	2 (L = 4)	3 (L = 8)	4 (L = 16)	5 (L = 32)	6 (L = 64)	7 (L = 128)	8 (L = 256)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	$\boxed{65,536}$	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Example: 128 ×**128 ×4 bits = 65536 bits**

Example 1

- A common measure of transmission for a digital data is the *Baud rate* (# of bits transmitted by second).

Generally, transmission is accomplished in packets consisting of a start bit, a byte (8 bits) of information, and a stop bit.

Example 1

- How many minutes it would take to transmit a 1024X1024 image with 256 gray levels, using a 56Kbaud modem?
- Total bits = $1024\times1024 \times[1+8+1]$= 1048576 bits
- Time required= 1048576/56000=187.25sec=3.1min.

- Repeat for a phone digital subscriber line (DSL) at 750K baud?
- Answer = 14 seconds.

Example 2

- A square image was represented by 131072 bits.
- What is the image spatial size if each pixel was represented by 8 bits?

Ans. N^2=Total bits/k

=131072/8=16384

N=√16384=128 pixels

Example 3

Consider an image of 512 x 512 dimension. Calculate the number of bits needs to store the file size of this image.

(a) If image is Binary.

(b) If image is Gray.

(c) If image is RGB.

Solution:

(a) Binary→ 1 Pixel= 1 Bit

$$512 \text{ X } 512 \text{ X } 1 \text{ Bit} = \frac{512 \ X \ 512}{8} \text{ Byte} = 32768 \text{ Byte} = \frac{32768}{1024} \text{ KB} = 32 \text{ KB} = \frac{32}{1024} \text{ MB} = .03125 \text{ MB}$$

(b) Gray→ 1 Pixel= 8 Bit

$$512 \text{ X } 512 \text{ X } 8 \text{ Bit} = \frac{512 \ X \ 512 \ X \ 8}{8} \text{ Byte} = 262144 \text{ Byte} = \frac{262144}{1024} \text{ KB} = 256 \text{ KB} = \frac{256}{1024} \text{ MB} = .25 \text{ MB}$$

(c) RGB→ 1 Pixel= 8 X 3 Bit

$$512 \text{ X } 512 \text{ X } 8 \text{ X } 3 \text{ Bit} = \frac{512 \ X \ 512 \ X \ 8 \ X \ 3}{8} \text{ Byte} = 786432 \text{ Byte} = \frac{786432}{1024} \text{ KB} = 768 \text{ KB} = \frac{768}{1024} \text{ MB} = .75 \text{ MB}$$

Aspect Ratio

- Proportional relation between image width and image height
- $Aspect\ Ratio = \frac{Width}{Height}$

- Example 1:
- If we want to resize a 1024x768 image to one that is 600 pixels wide with the same aspect ratio as the original image, what should be the height of the resized image?

(R X C = Height x Width)

Since $Aspect\ Ratio = \frac{Width}{Height}$

$$Therefore,\ Aspect\ Ratio = \frac{768}{1024} = 0.75$$

Also, Height$= \frac{Width}{Aspect\ Ratio} = \frac{600}{0.75} = 800$

- Example 2
- Suppose we have an image that have 20 rows and have an aspect ratio of 2:5. Find the total number of pixels in the image.

Since, $Aspect\ Ratio = \frac{Width}{Height}$

- $$\frac{2}{5} = \frac{Width}{20}$$

Width= 8

Therefore, Total number of Pixel = *R X C* = *Height x Width*

= 8 X 20 = 160 Pixel

Basic relationship between pixels

- •Neighborhood
- •Adjacency
- •Connectivity
- •Paths
- •Regions and boundaries

(i-1, j-1)	(i-1, j)	(i-1, j+1)
(i, j-1)	(i, j)	(i, j+1)
(i+1, j-1)	(i+1, j)	(i+1, j+1)

- 4-adjacency: Two pixels p and q with values from set 'V' are 4-adjacent if q is in the set $N_4(p)$
- Eg:
  - V = {0, 1}

1	1	0
1	1	0
1	0	1

- 8-adjacency: Two pixels p and q with values from set 'V' are 8-adjacent if q is in the set $N_8(p)$
- Eg:
  - V = {1, 2}

0	1	1
0	2	0
0	0	1

- Eg:
  - Find 4-adjacency and 8-adjacency of the center pixel
  - Note: V = {1}

Solution

- 4-adjacency

0	1	1
0	1	0
0	0	1

- Eg:
  - Find 4-adjacency and 8-adjacency of the center pixel
  - Note: V = {1}

Solution

- 8-adjacency

0	1	1
0	1	0
0	0	1

Basic relationship between pixels (Connectivity)

- Let S represent a subset of pixels in an image
- Two pixels p with coordinates (x_0, y_0) and q with coordinates (x_n, y_n) are said to be connected in S if there exists a path

$$(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$$

Basic relationship between pixels (Path)

- A path from pixel p with coordinates (x_0, y_0) to pixel q with coordinates (x_n, y_n) is a sequence of distinct pixels with coordinates $(x_0, y_0), (x_1, y_1), \ldots, (x_n, y_n)$
  - where (x_i, y_i) and (x_{i-1}, y_{i-1}) are adjacent for $1 \le i \le n$
- Here n is the length of the path
- If $(x_0, y_0) = (x_n, y_n)$, the path is closed path
- We can define 4-, 8-, and m-paths based on the type of adjacency used

Basic relationship between pixels (Path)

- Eg:
  - Compute the length of shortest-4 path between pixels p and q, where V = {1, 2}

Solution

	4	2	3	2	q
	3	3	1	3	
	2	3	2	2	
p	2	1	2	3	

x

Basic relationship between pixels (Path)

- Eg:
  - Compute the length of shortest-8 path between pixels p and q, where V = {1, 2}

Solution

	4	2	3	2	q
	3	3	1	3	
	2	3	2	2	
p	2	1	2	3	

Basic relationship between pixels (Path)

- Eg:
  - Compute the length of shortest-m path between pixels p and q, where V = {1, 2}

Solution

	4	2	3	2	q
	3	3	1	3	
	2	3	2	2	
p	2	1	2	3	

Distance Measure

- Given pixels p(u, v), q(w, x) and r(y, z) with coordinates, the distance function D has following properties:
  - $D(p, q) \geq 0$ [$D(p, q) = 0$, iff $p = q$]
  - $D(p, q) = D(q, p)$
  - $D(p, r) \leq D(p, q) + D(q, r)$

Distance Measure

- The following are the different distance measures
  - Euclidean Distance:
    - $D_e(p, q) = [(u-w)^2 + (v-x)^2]^{1/2}$
  - City Block Distance:
    - $D_4(p, q) = |u-w| + |v-x|$
  - Chess Board Distance:
    - $D_8(p, q) = \max(|u-w|, |v-x|)$

Pixel	Coordinate
p	(u, v)
q	(w, x)

Distance Measure (City-block Distance)

- The pixels having a D_4 distance from p(u, v) less than or equal to some value, form a diamond centered at p(u, v)
- The pixels with D_4 =1 are the 4-neighbors of p(u, v)

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

Distance Measure (Chessboard Distance)

- The pixels with D_8 distance from p(u, v) less than or equal to some value, form a square centered at p(u, v)
- The pixels with D_8=1 are the 8-neighbors of (u, v)

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2